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Analytics Vidhya

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Employment History

- July 2026 – Present **Assistant Professor**, Department of Computational Intelligence, SRM Institute of Science and Technology, Kattankullathur Campus
- 2019 – 2026 **Vellore Institute of Technology, Chennai Campus**, School of Electronics Engineering (SENSE)
- **Research Associate** (July 2019 – March 2020)
 - **Assistant Professor Junior** (March 2020 – August 2022)
 - **Assistant Professor (Contract)** (September 2022 – Present)
 - **Courses Delivered:** Problem Solving using Python, Computer Programming: Python, Computer Programming: Java, Embedded C Programming
- 2016 – 2019 **Assistant Professor**, Department of Electronics and Communication Engineering, GKM College of Engineering and Technology, Chennai
- 2013 – 2015 **Amrita Vishwa Vidyapeetham, Ettimadai**, Department of Electronics and Communication Engineering
- **Teaching Assistant** (2013 – 2014)
 - **TCS Research Scholar** (2014 – 2015)
- 2012 – 2013 **Assistant Professor**, Department of Biomedical Engineering, Vel Tech Multi Tech Dr.Rangarajan Dr.Sakunthala Engineering College, Avadi
- 2011 – 2012 **Assistant Professor**, Department of Electronics and Communication Engineering, VSB Engineering College, Karur

Education

- 2019 – Present **Ph.D., Artificial Intelligence and BioMedical Signal Analytics, Vellore Institute of Technology, Chennai campus** Thesis title: *Deep Learning Architecture for Effective Classification of ECG Waveforms* (Thesis Submitted)
- 2009 – 2011 **M.E. Power Electronics and Drives, Rajalakshmi Engineering College, Chennai** Thesis title: *Voltage Sag Mitigation of ASD*
- 2005 – 2009 **B.Tech. Electronics and Communication Engineering, Amrita Vishwa Vidyapeetham, Bangalore campus** Thesis title: *Tracking the Wild: A Sensor-Based Collar System for Monitoring Animal Movements in Forest Ecosystems*

Research Publications

Journal Articles

- 1 S. Premanand and S. Narayanan, "BlendNet: A blending-based convolutional neural network for effective deep learning of electrocardiogram signals," *Frontiers in Artificial Intelligence*, vol. 8, p. 1 625 637, Aug. 2025, Frontiers, SCI 4.7, Q1, ISSN: 2624-8212. DOI: 10.3389/frai.2025.1625637 Accessed: Aug. 24, 2025.

- 2 **S. Premanand** and S. Narayanan, "ECG-ResViT: A hybrid CNN-ViT model for efficient ECG signal classification," en, *The European Physical Journal Special Topics*, Jul. 2025, Publisher: Springer Science and Business Media LLC, SCI 2.3, Q2, ISSN: 1951-6355, 1951-6401.  DOI: 10.1140/epjs/s11734-025-01781-y Accessed: Jul. 21, 2025.
- 3 **P. S.** and S. Narayanan, "Convolution-based heterogeneous activation facility for effective machine learning of ecg signals," en, *Computers, Materials & Continua*, vol. 77, no. 1, pp. 25–45, 2023, SCI 3.1, Q2, ISSN: 1546-2226.  DOI: 10.32604/cmc.2023.042590 Accessed: Jun. 17, 2025.
- 4 **S. Premanand** and S. Narayanan, "A Tree Based Machine Learning Approach for PTB Diagnostic Dataset," *Journal of Physics: Conference Series*, vol. 2115, no. 1, p. 012 042, Nov. 2021, ISSN: 1742-6588, 1742-6596.  DOI: 10.1088/1742-6596/2115/1/012042 Accessed: Jun. 17, 2025.
- 5 **S. Premanand**, K. Vidya, D. Nivea, and T. GeethaPriya, "IMPROVED PERFORMANCE OF ASD UNDER VOLTAGE SAG CONDITIONS," *INTERNATIONAL JOURNAL OF ELECTRICAL ENGINEERING AND TECHNOLOGY (IJEET)*, vol. 4, no. 2, pp. 46–52, Mar. 2013, ISSN: 0976 – 6553.  URL: https://iaeme.com/Home/article_id/40220130402005
- 6 **S. Premanand**, A. Selvaraj, and S. R. Paranjothi, "Voltage Sag Mitigation on ASD," *International Journal of Engineering Science and Technology (IJEST)*, vol. 3, no. 2, pp. 1339–1343, Feb. 2011, ISSN: 0975-5462.

Books and Chapters

- 1 **S. Premanand** and S. Sugunavathy, "Augmenting the Performance of Multi-patient Parameter Monitoring System in LKSVM," in *Application of Computational Intelligence to Biology*, R. Bhramaramba and A. C. Sekhar, Eds., Series Title: SpringerBriefs in Applied Sciences and Technology, Singapore: Springer Singapore, 2016, pp. 9–18, ISBN: 978-981-10-0390-5 978-981-10-0391-2.  DOI: 10.1007/978-981-10-0391-2_2 Accessed: Jun. 17, 2025.
- 2 **S. Premanand** and S. Sugunavathy, "Enhancing the Performance of Multi-parameter Patient Monitors by Homogeneous Kernel Maps," in *Application of Computational Intelligence to Biology*, R. Bhramaramba and A. C. Sekhar, Eds., Series Title: SpringerBriefs in Applied Sciences and Technology, Singapore: Springer Singapore, 2016, pp. 1–7, ISBN: 978-981-10-0390-5 978-981-10-0391-2.  DOI: 10.1007/978-981-10-0391-2_1 Accessed: Jun. 17, 2025.
- 3 **S. Premanand**, C. Santhosh Kumar, and A. Anand Kumar, "Improving the Performance of Multi-parameter Patient Monitor System Using Additional Features," en, in *Computational Intelligence in Medical Informatics*, N. B. Muppalaneni and V. K. Gunjan, Eds., Series Title: SpringerBriefs in Applied Sciences and Technology, Singapore: Springer Singapore, 2015, pp. 53–58, ISBN: 978-981-287-259-3 978-981-287-260-9.  DOI: 10.1007/978-981-287-260-9_5 Accessed: Jun. 17, 2025.

Highlights & Endeavors

Academic Service & Peer Review Activities

-  Reviewer for *Scientific Reports* (Nature Portfolio)
-  Reviewer for *BMC Public Health* (Springer Nature)
-  Reviewer for *Knowledge and Information Systems* (Springer)
-  Reviewer for *Signal, Image and Video Processing* (Springer)

Awards and Achievements

- 2025  Honored with Raman Research Award twice for research work carried out at Vellore Institute of Technology, Chennai Campus.
- 2020  NPTEL – *Introduction to Machine Learning*, IIT Madras. Achieved a perfect score of **100%** (12-week course, Jan–Apr 2020), funded by the Ministry of HRD, Government of India (SWAYAM).
- 2019  Best Knowledge Transfer Staff Award for the contribution of Knowledge transfer gained by attending various programs during 2018-2019 in GKM College of Engineering and Technology, Chennai
- 2014  Selected in Research Scholar Program (TCS-RSP) Fellowship - Cycle 9, Amrita Vishwa Vidyapeetham, Ettimadai

Highlights & Endeavors

Guest Talks and Invited Lectures

- 2025
- Invited speaker in the IEEE Madras Section Sponsored Five-Day Faculty Development Programme on '**Generative AI, Large Language Models (LLMs), and Transformers**', organized by the Department of Information Technology, Sri Sairam Institute of Technology, Chennai (15–19 Dec 2025).
 - Delivered a guest lecture on '**AI-Driven Innovations in Electric Vehicles: From Data to Intelligent Decision-Making**', organized by the Department of Electrical and Electronics Engineering in association with the Centre of Excellence in Electric Vehicles, Rajalakshmi Engineering College, Chennai (21 Nov 2025).
 - Resource Person for the guest lecture on "**Digital Twin of Devices and Its Applications**" organized by the Department of Computer Science & Engineering (AI & ML) in association with IEEE EMBS SB Chapter, Sri Sairam Engineering College, Chennai, on 11-11-2025.
 - Resource Person for the workshop "**Learning ECG Intelligence: A Beginner's Journey into ML and DL for Cardiac Data**" at DRESTEIN '25, organized by the Department of ECE, Saveetha Engineering College on 05-11-2025.
 - Delivered a guest talk as a resource person on "**Personalized Medicine through AI and Data Analytics**" during a six-day STTP titled "*AI and Healthcare: Transforming the Future of Medicine*", organized by the Department of AI and DS, Panimalar Engineering College, Chennai (09.06.2025–14.06.2025).
 - Delivered an expert talk on "**Generative AI Image Generation (Hands-On)**", in an 8-day Virtual FDP (02.01.2025–10.02.2025), organized by the Department of AI and DS, EGS Pillay Engineering College, Nagapattinam.
 - Delivered a session titled "**Deep Learning in IoT: Enabling Real-Time Intelligent Systems**", during an FDP on "*IoT in Focus: Emerging Trends and Real-World Challenges*", organized by the Department of CSE (IoT), Sri SaiRam Engineering College, Chennai (22.01.2025).
 - Honoured as distinguished speaker on the topic "**Industry Applications, Career Path – Hands-On Image Classification and Text Generation**", in a six-day SDP on **Generative AI**, organized by the Department of AI and DS, EGS Pillay Engineering College, Nagapattinam (24.03.2025–29.03.2025).
- 2024
- Delivered a session on "**WEKA for Data Analysis**", organized by BANDVIT, TechnoVIT, VIT Chennai (18.09.2024).
 - Delivered a session on "**Computer Vision: Decoding the Visual World**", organized by BANDVIT and IEEE Signal Processing, TechnoVIT, VIT Chennai (12.09.2024).
 - Delivered a session on "**AIoT Unveiled: Integrating AI and IoT for a Smarter World**", organized by the Department of CSE-IoT, Sri SaiRam Engineering College, Chennai (24.07.2024).
 - Invited speaker in a five-day online FDP "**The Future of Generative AI in Academic Research and Publishing**", organized by the Department of CSE, Institute of Engineering and Management Group, Kolkata, sponsored by CSI Kolkata Chapter (06.01.2024).
- 2023
- Delivered a session on "**Navigating the Landscape of Modern Technologies and Data Science**", organized by the Department of CSE-IoT, Sri SaiRam Engineering College, Chennai (14.10.2023).
 - Invited as resource person for a two-day workshop on "**Machine Learning and Deep Learning Implementation**", organized by the Department of IT, M.Sc IT, M.O.P. Vaishnav College for Women (Autonomous), Chennai (11.09.2023–12.09.2023).
 - Delivered a session on "**The Art of Data Science: Transforming Data into Actionable Insights**", organized by the Department of CSE, Pollachi Institute of Engineering and Technology (17.02.2023).

Highlights & Endeavors (continued)

- 2022
- Delivered a guest lecture on “**Demystifying Machine Learning**”, organized by the Department of CSE, P.A. College of Engineering and Technology, Pollachi (17.11.2022).
 - Delivered a speech on “**IoT Technologies: The Present and the Future**”, during a CSIR-sponsored national seminar on “Advancements in IIoT for Engineering Applications,” organized by P.A. College of Engineering and Technology, Pollachi (04.08.2022).
 - Delivered a session on “**Innovation Design Using Machine Learning**”, organized by the Department of ECE, St. Mother Theresa Engineering College, Thoothukudi (30.03.2022).
 - Delivered a lecture on “**Machine Learning for Medical Data**”, during an AICTE-ISTE Induction/Refresher Program, organized by the Department of BME, Rajiv Gandhi College of Engineering and Technology, Puducherry (11.02.2022).
- 2020
- Delivered a webinar on “**Conceptual Elucidation of Machine Learning**” during SLAC 2020 Hackathon, Amrita School of Engineering, Bengaluru.

Short Term Courses Attended

- Attended a 5-day Online STTP on “**MATLAB-based Teaching-Learning in Mathematics, Science and Engineering**”, organized by Ramrao Adik Institute of Technology, Nerul, Navi Mumbai, in collaboration with DesignTech Systems Pvt. Ltd., Mumbai (18.05.2020–22.05.2020).
- 2019
- Participated in a 5-day short-term course on “**Data Science (Artificial Intelligence and Machine Learning)**”, organized by NIT Tiruchirappalli in association with eight industry partners (11.03.2019–15.03.2019).

Workshops Attended

- 2023
- Participated in a two-day online workshop on **Agri-tech 4.0 – The Future of Farming Technology**, organized by the Center of Advanced Data Science, VIT Chennai (06.05.2023–07.05.2023).
- 2019
- Attended a one-day workshop on **R Programming**, organized by the Training Learning Center, ICT at IIT Bombay, via A-View at VIT Chennai (09.11.2019).
 - Participated in a two-day workshop on **LaTeX**, organized by the School of Advanced Sciences (SAS), VIT Chennai (25.07.2019–26.07.2019).
- 2018
- Attended a three-day workshop on **Deep Learning**, organized by the Computational Intelligence Research Group, School of Computing Science & Engineering, VIT Chennai in association with Renaissance (01.02.2018–03.02.2018).
- 2015
- Participated in **TACTiCS – Innovation Lab Interactions on IoT**, organized under the TCS Research Scholarship Programme, at TCS Campus, Siruseri, Chennai (19.03.2015–21.03.2015).
- 2014
- Attended a workshop on **The Art of Technical Paper Writing and Professional Ethics**, sponsored by IETE, Amrita Vishwa Vidyapeetham, Coimbatore (Oct 2014).

Faculty Development Programmes Attended

- Participated in a two-week ISTE workshop on “**Signals and Systems**”, conducted by IIT Kharagpur under the National Mission on Education through ICT (MHRD, Govt. of India), hosted at Amrita Vishwa Vidyapeetham University, Coimbatore (Jan 2014).

Skills

Programming Languages & Tools:	C, Python, Java, WEKA, Git, GitHub, $\text{\LaTeX}$
Core Areas:	Statistics, Exploratory Data Analysis, Machine Learning, Deep Learning

References

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